Lab 3 Part 1 - Configuring a Zone-Based Policy Firewall (ZPF)

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
ZPF	G0/0	10.1.10.1	255.255.255.0	N/A
	G0/1	172.16.10.1	255.255.255.0	N/A
	G0/0/0	93.62.129.89	255.255.255.252	N/A
Web	NIC	10.1.10.100	255.255.255.0	10.1.10.1
Email	NIC	10.1.10.101	255.255.255.0	10.1.10.1
FTP	NIC	10.1.10.102	255.255.255.0	10.1.10.1
DHCP	NIC	172.16.10.100	255.255.255.0	172.16.10.1
Inside_PC1	NIC	172.16.10.10	255.255.255.0	172.16.10.1
DNS	NIC	8.8.8.8	255.255.255.240	8.8.8.1
Outside_PC	NIC	192.168.1.10	255.255.255.240	192.168.1.1

Objectives

- Verify connectivity among devices before firewall configuration.
- Configure a zone-based policy (ZPF) firewall on the ZPF router.
- Verify ZPF firewall functionality using ping, FTP, and a web browser.

Background/Scenario

A Zone-based policy (ZPF) firewall is an advanced method of the stateful firewall which changes the firewall from the older interface-based model to a more flexible, more easily understood zone-based configuration model. In this activity, you will configure a basic ZPF on an edge router that allows internal and external hosts controlled access to applications running on servers in the DMZ. You will also configure the ZPF to allow internal hosts access to external resources while blocking external hosts and DMZ hosts from accessing internal resources. You will then verify firewall functionality between the different zones.

The router has been pre-configured with the following:

- Host names and IP addressing
- Local username and password: **admin1** / **netsec123**
- Static routing

Part 1: Verify Basic Network Connectivity

Verify network connectivity prior to configuring the zone-based policy firewall.

Step 1: From the Outside_PC command prompt, ping Inside_PC1 at 172.16.10.10 and the FTP server at 10.1.10.102.

Step 2: Access the FTP server.

a. From the **Outside_PC** command prompt, establish an FTP session to the FTP server at **10.1.10.102**. Use the username **admin1** and password **netsec123** to log in.

```
C:\> ftp 10.1.10.102
Trying to connect...10.1.10.102
Connected to 10.1.10.102
220- Welcome to PT Ftp server
Username: admin1
331- Username ok, need password
Password: <netsec123>
230- Logged in
(passive mode On)
ftp>
```

b. Exit the FTP session using **quit**.

Step 3: From Outside_PC, open a web browser to the Web server.

a. Click the **Desktop** tab and then click the **Web Browser** application. Enter the **netseclab3.ie** as the URL. The welcome page from the web server should be displayed.

b. Close the browser on Outside_PC.

QUESTION 1 Which of the connection types (if any) were successful? (Choose all that apply)

Before proceeding, go to the "Lab 3 Part 1 - Configuring a Zone-Based Policy Firewall (ZPF) - QUESTIONS" quiz on the Brightspace page and enter your answer for question 1. Leave the quiz open while you complete the rest of the lab sheet.

Part 2: Create the Firewall Zones on ZPF

Note: For all configuration tasks, be sure to use the exact names as specified.

Step 1: Create an internal zone.

Use the **zone security** command to create a zone named **INSIDE**.

```
ZPF(config)# zone security INSIDE
ZPF(config-sec-zone) exit
```

Step 2: Create an external zone.

Use the **zone security** command to create a zone named **OUTSIDE**.

Step 3: Create a DMZ zone.

Create a zone named **DMZ**.

Part 3: Identify DMZ Traffic Using a Class-Map

In this part you will define a class map that contains all of the protocols that are permitted to the DMZ zone. The servers in the DMZ should accept requests using Web (HTTP & HTTPS), Email (SMTP & POP3) or FTP.

Create a class map named **DMZ-TRAFFIC** using the **class-map type inspect** command with the **match-any** option to. Use the **match protocol** command to each type of protocol allowed.

```
ZPF(config)# class-map type inspect match-any DMZ-TRAFFIC
ZPF(config-cmap)# match protocol http
ZPF(config-cmap)# match protocol https
ZPF(config-cmap)# match protocol smtp
ZPF(config-cmap)# match protocol pop3
ZPF(config-cmap)# match protocol ftp
ZPF(config-cmap)# exit
```

Part 4: Specify Firewall Policies for Traffic Between the Outside and DMZ

In this part you will create a policy map to specify what happens to the class of traffic previously defined. The available actions are to *pass*, *drop* and *inspect* the traffic.

Step 1: Create a policy map to determine what to do with matched traffic.

Create a policy map named **OUT-2-DMZ-PMAP** using the **policy-map type inspect** command.

```
ZPF(config)# policy-map type inspect OUT-2-DMZ-PMAP
```

Step 2: Specify a class type of inspect and reference class map DMZ-TRAFFIC.

Associate the class **DMZ-TRAFFIC** configured in part 3 with the policy-map using the **class type inspect** command.

```
ZPF(config-pmap)# class type inspect DMZ-TRAFFIC
```

Step 3: Specify the action of inspect for this policy map.

Use the **inspect** action to configure the ZPF to maintain state information for all traffic that is a member of the class DMZ-TRAFFIC.

```
ZPF(config-pmap-c)# inspect
```

Issue the **exit** command twice to leave **config-pmap-c** mode and return to **config** mode.

```
ZPF(config-pmap-c)# exit
ZPF(config-pmap)# exit
```

Part 5: Apply Firewall Policies

In this part you will define a zone pair between the OUTSIDE zone and the DMZ zone and associate it with the policy map just created. You will then assign the zones to their applicable interfaces.

This will apply the action of inspect (as specified in our policy map) to any traffic type defined in our class map when going from the OUTSIDE to the DMZ

Step 1: Create a pair of zones.

Create a zone pair named **OUT-2-DMZ-ZPAIR** using the **zone-pair security** command. Specify the source as the **OUTSIDE** zone and the destination as the **DMZ** zone which we created in Part 3.

```
ZPF(config)# zone-pair security OUT-2-DMZ-ZPAIR source OUTSIDE destination DMZ
```

Step 2: Specify the policy map for handling the traffic between the two zones.

Attach a policy-map and its associated actions to the zone pair using the **service-policy type inspect** command and reference the policy map previously created, **OUT-2-DMZ-PMAP**.

```
ZPF(config-sec-zone-pair)# service-policy type inspect OUT-2-DMZ-PMAP
ZPF(config-sec-zone-pair)# exit
ZPF(config)#
```

Step 3: Assign interfaces to the appropriate security zones.

Use the **zone-member security** command in interface configuration mode to assign G0/0/0 to **OUTSIDE** and G0/0 to **DMZ.**

```
ZPF(config)# interface g0/0/0
ZPF(config-if)# zone-member security OUTSIDE
ZPF(config-if)# exit
ZPF(config)# interface g0/0
ZPF(config-if)# zone-member security DMZ
ZPF(config-if)# exit
```

Step 4: Copy the running configuration to the startup configuration.

Part 6: Test Firewall Functionality from OUTSIDE to DMZ

Verify that outside hosts can still access services in the DMZ after configuring the ZPF.

Step 1: From external Outside_PC, ping the FTP server in the DMZ.

From the **Outside_PC** command prompt, ping **FTP** at 10.1.10.102.

QUESTION 2 Was the Outside_PC still able to ping the FTP server? If not, why was this the case?

Step 2: From external Outside_PC, FTP to the FTP server in the DMZ.

a. From the **Outside_PC** command prompt, FTP to the **FTP** server at 10.1.10.102. Use the username **admin1** and the password **netsec123**. The FTP session should succeed.

b. While the FTP session is active, issue the command **show policy-map type inspect zone-pair sessions** on **ZPF** to view established sessions.

QUESTION 3 What is the source IP address?

QUESTION 4 What is the destination IP address?

QUESTION 5 What is the destination port number?

Before proceeding, return to the quiz on the Brightspace page and enter your answers for questions 3-5. Leave the quiz open while you complete the rest of the lab sheet.

Step 3: From Outside_PC, exit the FTP session by entering quit and close the command prompt window.

Step 4: From Outside_PC, open a web browser to the Web server web page.

Enter **netseclab3.ie** in the browser URL field, and click **Go**. The HTTP session should succeed. While the HTTP session is active, issue the command **show policy-map type inspect zone-pair sessions** on **ZPF** to view established sessions.

Note: If the HTTP session times out before you execute the command on **ZPF**, you will have to click the **Go** button on **Outside_PC** to generate a session between **Outside_PC** and **Web**.

QUESTION 6 What is the source IP address?

QUESTION 7 What is the destination IP address?

QUESTION 8 What is the destination port number?

Before proceeding, return to the quiz on the Brightspace page and enter your answers for questions 6-8. Leave the quiz open while you complete the rest of the lab sheet.

Step 5: Close the browser on Outside_PC.

Part 7: Test Firewall Functionality Between Other Zones

In this part you will verify that external hosts cannot access internal resources after configuring the ZPF. Also, you will check if internal hosts can access the DMZ.

Step 1: From Outside_PC's command prompt, ping Inside_PC1.

From the **Outside_PC** command prompt, ping **Inside_PC1** at 172.16.10.10. The ping should fail.

QUESTION 9 Why was the ping unsuccessful?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 9. Leave the quiz open while you complete the rest of the lab sheet.

Step 2: From Inside_PC1, open a web browser to the Web server web page.

Enter **netseclab3.ie** in the browser URL field of Inside_PC1, and click **Go**. The HTTP session should fail.

Part 8: Create a Zone Pair Between INSIDE and DMZ Using the Existing Firewall Policy

One of the advantages of ZPFs is that class maps and policy maps can be reused between different zone pairs as needed. In this part you will define a new zone pair between the INSIDE zone and the DMZ zone and associate it with the policy map you already created. You will then assign the INSIDE zone to its applicable interface.

This will also apply the action of inspect (as specified in our policy map) to any traffic type defined in our class map when going from the INSIDE to the DMZ

Step 1: Create a pair of zones.

Create a zone pair named **IN-2-DMZ-ZPAIR**. Specify the source as the **INSIDE** zone and the destination as the **DMZ** zone.

Step 2: Specify the policy map for handling the traffic between the two zones.

Attach the policy map previously created, **OUT-2-DMZ-PMAP**, to the zone pair.

Step 3: Assign the G0/1 interface to the INSIDE zone.

Assign G0/1 to the **INSIDE** zone.

*Note: We already assigned G0/0 to the **DMZ** zone so no need to do so again.*

Step 4: Test firewall functionality from INSIDE to DMZ.

a. From **Inside_PC1**, FTP to the **FTP** server in the DMZ.

b. From **Inside_PC1**, test connectivity to the **Web** server by opening the web browser, entering **netseclab3.ie** in the browser URL field, and clicking **Go**.

c. From **Inside_PC1**, test connectivity to the **Web** server by opening the web browser, entering **10.1.10.100** in the browser URL field, and clicking **Go**.

QUESTION 10 Which, if any, of the connection tests from Inside_PC1 to the DMZ zone were successful?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 10. Leave the quiz open while you complete the rest of the lab sheet.

Part 9: Implement a Policy for Traffic Between the INSIDE and OUTSIDE

While we have already created policies for traffic going to the DMZ zone from both the INSIDE and OUTSIDE zones, we have not yet created a policy to allow traffic to go from the INSIDE zone to the OUTSIDE zone. As the DNS server is currently in the OUTSIDE zone, in this part we will create a policy to allow DNS requests out of our internal network to the DNS server on the Internet. We will also allow ICMP traffic for diagnostic purposes.

Hint: For help with the commands for this part, look back over what we did previously in Parts 3-5 (and/or lecture 3).

Step 1: Identify traffic allowed from INSIDE to OUTSIDE using a Class-Map.

Create a class map named **OUTGOING-TRAFFIC** which will match any of the **ICMP** or **DNS** protocols.

Step 2: Create a policy map to determine what to do with matched traffic between the INSIDE and OUTSIDE.

Create a policy map named **IN-2-OUT-PMAP** and configure it to **inspect** traffic defined in the **OUTGOING-TRAFFIC** class map created in the previous step.

Step 3: Create a pair of zones and attach the policy map to it.

Create a zone pair named **IN-2-OUT-ZPAIR** specifying the source as the **INSIDE** zone and the destination as the **OUTSIDE** zone. Attach the **IN-2-OUT-PMAP** policy map to the zone pair.

Step 4: Test firewall functionality from INSIDE to OUTSIDE.

a. From **Inside_PC1**, ping **Outside_PC**.

b. From **Outside_PC**, ping **Inside_PC**.

c. From **Inside_PC1**, test connectivity to the **Web** server by opening the web browser, entering **netseclab3.ie** in the browser URL field, and clicking **Go**.

QUESTION 11 Which, if any, of the connection tests involving the INSIDE zone were successful this time?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 11.

If you have correctly configured all parts of the lab your activity score should now be showing as 100%. If so, click on "check results" in the activity window. Return to the Brightspace quiz one last time and enter the code into the appropriate question box (Q12) of the quiz.

You have completed the lab - please submit the Brightspace quiz.